

Functional Requirements Document (FRD)

1. Scenario Simulator & Live Predict

FR-1.1: The system must allow users to input Country Name, Life Expectancy, Expected Schooling, Mean Schooling, and GNI per Capita.

FR-1.2: Sliders must sync bidirectionally with numeric input fields in real-time.

FR-1.3: Triggering predictions must run asynchronous AJAX calls, returning predictions without page reloads.

2. Explainable AI & Advisory Recommendations

FR-2.1: The prediction output must display a circular confidence gauge and an explanation paragraph describing the prediction drivers.

FR-2.2: The system must list positive and negative contributing indicators based on input values.

FR-2.3: The system must output actionable policy recommendations for lagging indicators.

3. Dataset Explorer & History Logs

FR-3.1: The dataset page must display historical indicators with pagination and category search filters.

FR-3.2: The history log must allow users to view, search, sort, and delete past prediction records.

FR-3.3: Users must be able to export filtered history tables as a formatted CSV file.